

William Liu

<https://renaissancedog.github.io> | jiyangliu729@gmail.com | San Antonio, Texas

Motivated high school senior with strong problem-solving ability, leadership experience, and a passion for STEM. Enjoys solving challenging problems, building creative projects, exploring advanced maths and sciences, and promoting STEM within the community. Aspiring to become a scientist or engineer to help create a better shared future.

Education

Ronald Reagan High School - 12th grade, expected graduation May 2027

- **GPA/Test Scores:** 99.2/100, unweighted GPA; 108.0/100 weighted GPA; 1600, SAT; 6, AIME; 108, AMC 12
- **Relevant Coursework:** AP Calculus BC, AP Physics C: Mechanics, AP Chemistry, AP Statistics

Experience

Research Assistant, Henkelman Research Group

February 2026 - Present

- Worked with a chemistry research group at UT Austin to study novel materials for use in lithium-ion batteries
- Used advanced molecular simulation libraries to write code and perform computations for different materials

Independent Developer

August 2023 - Present

- Developed several projects to explore programming, mathematics, hardware systems, and data analysis
- Built a basketball data analysis program with Python and SQLite to analyze video game statistics and project player performance with predictive modeling
- Developed several interactive web applets, including a Valentine's Day website and a bell schedule utility
- Designed and prototyped a functional alarm clock using Arduino, breadboard circuitry, and embedded C++
- Created a customized personal website with Svelte to showcase portfolio of projects and technical skills

Reagan Mu Alpha Theta

VP of Academics (April 2026 - Present), Inducted Member (October 2023 - Present)

- Organized and directed math contests, study events, and games for school chapter of national math honor society, reaching over 50 students
- Developing Python-based administrative automation tools for chapter operations, and leading efforts to streamline and restructure club operations

Reagan Orchestra

Librarian (May 2025 - Present), Violinist (August 2023 - Present)

- Auditioned into varsity Chamber Orchestra and All-Region Orchestra multiple times
- Managed copying, inventorying, organizing, and distributing music as a Librarian

Skills

- **Technical:** Strong mathematical/programming skills, knowledge of AI/ML concepts, AutoCAD Certified User
- **Programming Languages:** Python, Java, JavaScript, HTML/CSS, SQLite
- **Programming Frameworks:** Pandas, NumPy, PyTorch, Svelte, matplotlib, seaborn, TailwindCSS
- **Other Interests:** Playing violin, running, swimming, reading, dreaming about public transit, watching NFL games

Awards

- American Invitational Mathematics Examination (AIME) Qualifier (2024, 2025, 2026): Only qualifier from school, first freshman qualifier in school history
- American Mathematics Competitions (AMC) 1st Place, Reagan High School (2023, 2024, 2025)
- USA Computing Olympiad (USACO) Silver Division
- Student Volunteer Service Award - Gold Level (2025, 2026)
- Texas State Solo & Ensemble Competition qualifier (2024-2026)
- Stanford Math Tournament 5th place Team Round with the San Antonio Math Circle (2026)
- US National Chemistry Olympiad qualifier (2026)